SQL TECHNICAL DOCUMENTATION

1. Project Overview

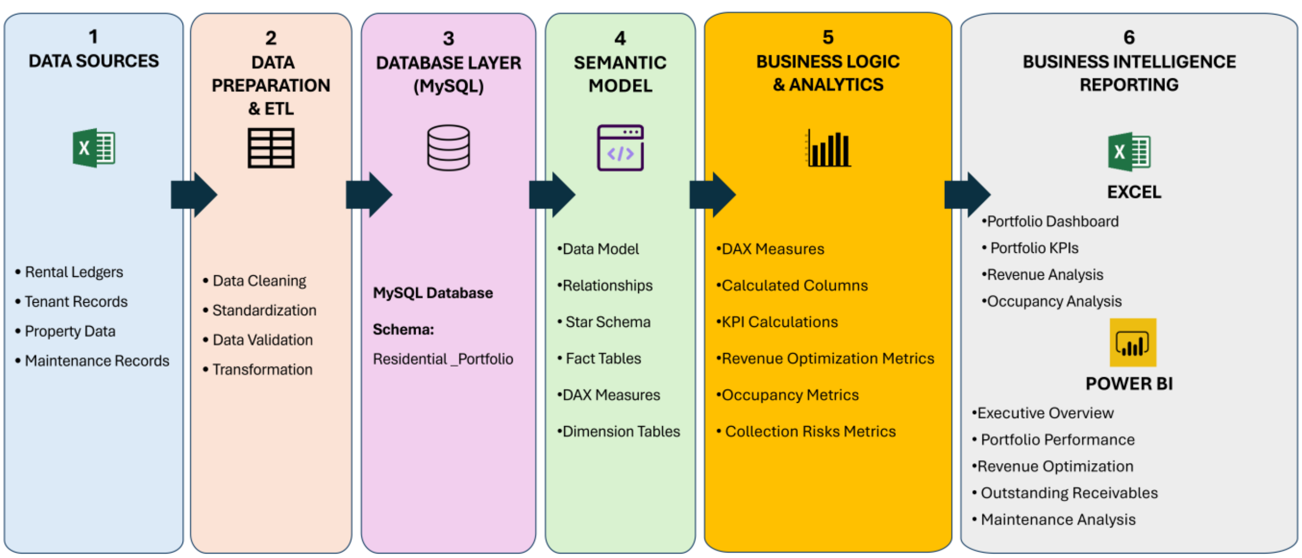
A MySQL relational database was developed to support the analysis of a residential property management portfolio. The database consolidates tenant, unit, maintenance, payment, and rent optimization data into a centralized analytical model for reporting and business intelligence.

SQL was used for:

- Database Design
- Data Storage
- Data Staging & Cleaning
- Feature Engineering
- Analytical View Creation
- Business Intelligence & Dashboard Reporting

2. Data Flow

Residential Portfolio Analytics
End-to-End Business Intelligence Architecture



Purpose: This architecture illustrates the end-to-end Business Intelligence workflow used to transform raw residential property data into interactive dashboards and actionable insights using MySQL, Excel, and Power BI.

2.1 Database Structure

OBJECT	NAME	DESCRIPTION
Database	residential_portfolio	Main analytical database

2.2 Database Objects Overview

The database contains six core tables and multiple analytical views that support reporting, business intelligence, and portfolio analysis.

TABLES		
Table	Tenants	Tenant and lease information
Table	Units	Property and apartment information
Table	Maintenance	Maintenance history and repair costs
Table	Detailed_Ledger	Tenant payment history
Table	Buildings	Building-level property information
Table	Rent_Optimization	Rent recommendation calculations
VIEWS		
View	vw_units	Consolidated unit reporting view
View	vw_average_rent	Average rent statistics
View	vw_portfolio_summary	Portfolio-wide KPIs and executive reporting
View	vw_executive_dashboard	Executive reporting dashboard
View	vw_low_rent_opportunities	Revenue optimization opportunities
View	vw_regulation_summary	Regulated vs. market-rate analysis
View	vw_maintenance_summary	Maintenance costs, repairs, and asset management.
View	vw_high_maintenance_units	High-cost maintenance units
View	vw_arrears_summary	Tenant arrears summary
View	vw_receivables_aging	Tenant balances and receivable monitoring.
View	vw_delinquency_opportunity	Delinquency opportunity analysis
View	vw_top_delinquent_tenants	Highest delinquent tenants
View	vw_top_delinquent_tenants_regulation	Identifies the highest delinquent tenants and categorizes outstanding balances by regulation status
View	vw_unit_level_outstanding_receivables	Summarizes outstanding receivables at the individual unit level

2.3 Data Dictionary

Table: Tenants

COLUMN	DATA TYPE	DESCRIPTION
Tenant_ID	VARCHAR(50)	Unique identifier for each tenant (Primary Key).
Move_In_Date	DATE	Date the tenant moved into the unit.
Tenant_Status	VARCHAR(30)	Current tenant status (e.g., Active, Vacated).
Lease_Start_Date	DATE	Lease commencement date.
Lease_End_Date	DATE	Lease expiration date.
Lease_Duration	INT	Lease length in months.
Lease_Type	VARCHAR(50)	Lease classification.
Occupancy_Years	INT	Number of years the tenant has occupied the unit.
Tenant_Segment	VARCHAR(50)	Tenant segmentation category.
Current_Rent	DECIMAL(10,2)	Current monthly rent charged.
Legal_Rent	DECIMAL(10,2)	Maximum legal rent (regulated units).
Prior_Year_Rent	DECIMAL(10,2)	Previous year's monthly rent.
Employment_Status	VARCHAR(30)	Tenant employment status.
Income	DECIMAL(12,2)	Annual household income.
Income_Category	VARCHAR(50)	Income classification used in analysis.
Bucket	VARCHAR(20)	Analytical grouping category.

TABLE: MAINTENANCE

COLUMN	DATA TYPE	DESCRIPTION
Repair_ID	INT	Unique repair identifier (Primary Key).
Tenant_ID	VARCHAR(50)	Tenant associated with the repair.
Unit_ID	VARCHAR(30)	Apartment requiring maintenance.
Units_Status	VARCHAR(20)	Unit occupancy status.
Tenant_Status	VARCHAR(30)	Tenant occupancy status.
Repair_Type	VARCHAR(50)	Type of maintenance performed.
Repair_Date	DATE	Date of repair.
Repair_Cost	DECIMAL(10,2)	Cost of repair.
Estimated_Savings	DECIMAL(10,2)	Estimated savings generated.
Priority_Level	VARCHAR(20)	Maintenance priority.
Annualized_Savings	DECIMAL(10,2)	Annual projected savings.
Repair_ROI	DECIMAL(10,3)	Return on investment for the repair.
Useful_Life_Years	INT	Expected useful life after repair.

TABLE: RENT_OPTIMIZATION

Column	Data Type	Description
Tenant_ID	VARCHAR(50)	Tenant identifier.
Unit_ID	VARCHAR(30)	Apartment identifier.
Current_Rent	DECIMAL(10,2)	Current monthly rent.
Regulation_Status	VARCHAR(30)	Market or regulated status.
Income_Category	VARCHAR(50)	Tenant income classification.
Occupancy_Years	INT	Years occupied.
Annual_Repair_Cost	DECIMAL(10,2)	Annual maintenance cost.
Maintenance_Ratio	DECIMAL(10,4)	Maintenance cost relative to rent.
Low_Rent_Flag	INT	Indicates below-market rent.
Long_Tenure_Flag	INT	Indicates long-term tenancy.
Income_Adjustment	VARCHAR(50)	Income adjustment category.
Low_Rent_Adjustment	DECIMAL(10,4)	Rent adjustment factor.
Long_Tenure_Adjustment	DECIMAL(10,4)	Long tenancy adjustment factor.
Tenancy_Companion	VARCHAR(30)	Tenancy classification.
FM_Turnover_Risk_Adjustment	VARCHAR(10)	Market turnover adjustment.
Recommended_Increase_Percentage	DECIMAL(10,4)	Suggested rent increase.
New_Rent	DECIMAL(10,2)	Recommended monthly rent.
Monthly_Increase	DECIMAL(10,2)	Monthly increase amount.
Annual_Revenue_Impact	DECIMAL(10,2)	Annual revenue gain.
Increase_Bucket	VARCHAR(20)	Increase category.

TABLE: DETAILED_LEDGER

Column	Data Type	Description
Payment_ID	VARCHAR(80)	Unique payment transaction identifier (Primary Key).
Tenant_ID	VARCHAR(50)	Tenant identifier.
Unit_ID	VARCHAR(30)	Apartment identifier.
Ledger_Year	INT	Financial year.
Month_Number	INT	Month number.
Month_Name	VARCHAR(20)	Calendar month.
Due_Date	DATE	Payment due date.
Prior_2022_Balance	DECIMAL(10,2)	Opening balance.
Amount_Due	DECIMAL(10,2)	Rent due.
Amount_Paid	DECIMAL(10,2)	Rent received.
Past_Due_Balance	DECIMAL(10,2)	Outstanding balance.
Payment_Date	DATE	Date payment received.
Days_Past_Due	INT	Number of days overdue.
Is_Late_After_Grace	INT	Late payment flag.
Payment_Status	VARCHAR(30)	Current payment status.

TABLE: UNITS

Column	Data Type	Description
Unit_ID	VARCHAR(30)	Unique apartment identifier (Primary Key).
Building_ID	VARCHAR(20)	Building identifier (Foreign Key).
Unit_Number	VARCHAR(20)	Apartment number.
Regulation_Status	VARCHAR(30)	Market or regulated.
Bedrooms	INT	Number of bedrooms.
Bathrooms	DECIMAL(3,1)	Number of bathrooms.
Washer_Dryer	VARCHAR(10)	Washer and dryer availability.
Dishwasher	VARCHAR(10)	Dishwasher availability.
Renovated	VARCHAR(10)	Renovation status.
Floor	INT	Floor number.
Occupied_Status	VARCHAR(20)	Occupancy status.

3. SQL Views

The Property Management database contains multiple analytical views designed to separate transactional data from reporting and business intelligence. Each view supports a specific area of portfolio analysis and can be reused by reporting tools such as Power BI and Excel.

CATEGORY	VIEWS	PURPOSE
Portfolio Reporting	vw_portfolio_summary, vw_executive_dashboard, vw_average_rent	Portfolio-level KPIs and executive reporting
Rent Analysis	vw_low_rent_opportunities, vw_regulation_summary	Rent optimization and regulated unit analysis
Maintenance	vw_maintenance_summary, vw_high_maintenance_units	Maintenance costs, repairs, and asset management
Receivables	vw_arrears_summary, vw_receivables_aging, vw_unit_level_outstanding_receivables	Tenant balances and receivable monitoring
Delinquency	vw_top_delinquent_tenants, vw_top_delinquent_tenants_regulation, vw_delinquency_opportunity	Collection prioritization and delinquency analysis
Operational	vw_units	Consolidated unit-level reporting

4. Feature Engineering

Several metrics were calculated within SQL before importing the data into Power BI.

METRIC	HOW IT IS CALCULATED / USED
Occupancy Rate	Measures the percentage of occupied units in the portfolio. Calculated as occupied units divided by total units.
Vacancy Count	Measures the number of unoccupied units. Calculated as total units minus occupied units.
Average Rent	Calculates the average monthly rent across all occupied units.
Monthly Increase	Measures the difference between the recommended rent and the current rent.

Annual Revenue Impact	Measures the projected annual revenue gain from rent optimization.
Maintenance Ratio	Compares maintenance cost against rental income to identify high-cost units.
Repair ROI	Compares estimated savings against repair cost to evaluate maintenance investment value.
Total Arrears	Calculates the total outstanding tenant balances across the portfolio.
Receivables Aging	Groups unpaid balances by how long they have been outstanding.
Days Past Due	Measures how many days a payment is overdue.
Rent-to-Income Ratio	Compares tenant rent against income to evaluate affordability.
Low Rent Flag	Identifies units with below-market or unusually low rent.
Long Tenure Flag	Identifies long-term tenants for rent optimization analysis.
Total Maintenance Cost	Measures total repair and maintenance spending across the portfolio.
Annual Repair Cost	Measures yearly maintenance costs at the unit or portfolio level.
Prior 2022 Arrears	Identifies outstanding balances carried forward from 2022.
Revenue Opportunity	Calculates the total potential annual revenue increase resulting from recommended rent adjustments.
Current Annual Revenue	Estimates current yearly rental income based on existing monthly rents.
Projected Annual Revenue	Estimated annual rental revenue after applying the recommended rent increases

5. Analytical Queries

Business-focused SQL queries were developed to support operational reporting and strategic decision-making, including:

- Vacancy analysis
- Delinquent tenant identification
- Revenue opportunity analysis
- Rent affordability assessment
- Maintenance cost analysis

- Portfolio risk assessment
- Executive portfolio reporting

6. SQL Skills Demonstrated

- Relational Database Design
- Table Creation
- Primary & Foreign Keys
- Data Staging & Cleaning
- Joins
- CASE Statements
- Aggregate Functions
- Window Functions
- CTEs
- Subqueries
- Views
- SQL Views Development
- Business KPI Development
- Aggregate Reporting
- NULL Handling
- Business Intelligence Data Preparation

7. Business Value

The SQL database provides a centralized analytical platform for residential property management. By integrating tenant, maintenance, receivables, and rent optimization data into reusable analytical views, the solution enables consistent reporting, portfolio monitoring, revenue optimization, maintenance planning, and executive decision-making while serving as the data source for future Power BI dashboards.

8. Database Summary

ITEM	VALUE
Database	1
Primary Tables	6
Analytical Views	14
Business Analysis Queries	7
Reporting Platform	MySQL / Power BI Ready